

Applications of Differential Calculus (Related Rates Part II) Homework

Name
Date
Course

$$V = \frac{4}{3}\pi r^3$$

- ① A spherical balloon is inflated. At what rate is its volume increasing when its radius is 6 in and the rate that its radius is increasing is 2 in/s?

$$A = 6s^2$$

- ② The surface area of a cube (i.e. the area of the outside surfaces of the cube) is increasing by $3 \text{ cm}^2/\text{h}$. How fast are the edge lengths of the cube increasing when the edge lengths are 10 cm?

$$A = 4\pi r^2$$

③ A snowball is melting. If the surface area of the ball decreases by $3 \text{ ft}^2/\text{min}$ when the radius is 1.1 ft , at what rate is the radius of the ball decreasing?

$$V = \frac{1}{3}\pi r^2 h$$

④ A water tank has the shape of a circular cone and the tip of the cone is the bottom of the tank. The radius of the top of the tank is 3 yd and the height of the tank is 9 yd . If 5 yd^3 of water are poured into the tank per minute, find the rate that the water level is rising when the water is 6 yd deep.

$$V = \frac{1}{3}\pi r^2 h$$

⑤ Sand is poured onto the ground creating a pile in the shape of a circular cone. The height of the pile is always half the diameter of its base. If the diameter of its base is increasing by 3.4 m/h, what volume of sand is being poured per hour when the radius of the base of the pile is 2 m?

⑥ A man 4.5 ft tall walks away from a street light at a speed of 3 ft/s. The light is 14 ft above the ground. At what rate is the length of his shadow increasing?

⑦ A light on the ground shines on a wall 16 ft away. If a woman 4 ft tall walks from the light to the wall at a speed of 2 ft/s, how fast is the length of her shadow on the wall decreasing when she is 4 ft from the wall?

⑧ How fast is the tip of the shadow moving in problem #6?

Answers

- ① $288\pi \text{ in}^3/\text{s}$ or $904.78 \text{ in}^3/\text{s}$ ⑤ $\frac{34\pi}{5} \text{ m}^3/\text{h}$ or $21.363 \text{ m}^3/\text{h}$
- ② $\frac{1}{40} \text{ cm/h}$ or 0.025 cm/h ⑥ $\frac{27}{19} \text{ ft/s}$ or $1\frac{8}{19} \text{ ft/s}$
- ③ $-\frac{15}{44\pi} \text{ ft/min}$ or -0.1085 ft/min ⑦ $-\frac{8}{9} \text{ ft/s}$
- ④ $\frac{5}{4\pi} \text{ yd/min}$ or 0.3979 yd/min ⑧ $\frac{84}{19} \text{ ft/s}$ or $4\frac{8}{19} \text{ ft/s}$

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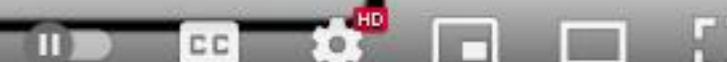
Calculus I

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Calculus I Course--Class 19: Applications of Differential Calculus (Related Rates Part II)

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